Junhong Zhang

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Shenzhen University Computer Science •PhD candidate Shenzhen

Educational Background

2022.09 to Present	College of Computer Science and Software Engineering, **Shenzhen University** Doctor (Direct PhD) in Computer Science and Technology
2018.09 to 2022.06	College of Computer Science and Software Engineering, **Shenzhen University** Bachelor; GPA: 4.21/4.50 (Rank 1 in major) in Computer Science and Technology (dual-degree program in Math and Computing Science).

Publications

Research focus: optimization, large-scale learning, representation learning. Published 14 papers (5 first-author: 4 in CAS Zone 1, 2 in CCF-A venues, and the rest are collaborative). Selected publications:

2025.04 — **J. Zhang**, Z. Lai, H. Kong, and J. Yang, “Learning The Optimal Discriminant SVM with Feature Extraction,” IEEE Trans. Pattern Anal. Mach. Intell. (T-PAMI), vol. 47, no. 4, pp. 2897-2911, Apr. 2025. **(CCF-A)**

- An adaptive joint learning model of features and classification with complete convergence analysis and efficient engineering implementation.
- Project: `https://github.com/Apple-Zhang/ODSVM-paper`

2025.05 — **J. Zhang** and Z. Lai, “Joker: Joint Optimization Framework for Lightweight Kernel Machines,” in Proc. 42nd Int. Conf. Mach. Learn. (ICML), 2025. **(CCF-A)**

- A general nonlinear learning framework for large-scale data: processes 10M data points with a single consumer-grade GPU of 10GB memory.
- Project: `https://github.com/Apple-Zhang/Joker-paper`

2026.02 — **J. Zhang**, Z. Lai, J. Zhou, G. Liang, “NPSVC++: A Representation Learning Framework for Nonparallel Classifiers,” in IEEE Trans. Neural Networks Learn. Syst. (T-NNLS), 2026. **(CAS Zone 1)**

- A Pareto-optimal representation learning framework for nonparallel classifiers, improving classification accuracy by 1.5% over conventional cross-entropy.
- Project: `https://github.com/Apple-Zhang/NPSVCPP`

2025.11 — **J. Zhang** and Z. Lai, “LighTwSVM: Efficient Linear Nonparallel Classifier for Millions of Data,” Pattern Recognition (PR), Art. no. 112700, 2025. **(CAS Zone 1)**

- An efficient drop-in replacement for `sklearn.svm.LinearSVC`, about 2x faster than LIBLINEAR with roughly 50% lower memory usage.
- Project: `https://github.com/Apple-Zhang/LighTwSVM-paper`

2023.04 — **J. Zhang**, Z. Lai, H. Kong, and L. Shen, “Robust Twin Bounded Support Vector Classifier with Manifold Regularization,” IEEE Trans. Cybern. (T-CYB), vol. 53, no. 8, pp. 5135-5150, 2023. **(CAS Zone 1)**

- A nonparallel classifier integrating a nonconvex robust metric and manifold structure to suppress noise, with global convergence guarantee.
- Project: `https://github.com/Apple-Zhang/RMTBSVM-paper`

2026.04 — F. Chen, **J. Zhang**, Z. Lai, and H. Kong, “TaylorNet: Rethinking Monomial-based Graph Neural Networks with Taylor Expansion,” Pattern Recognition (PR), vol. 172, Art. no. 112748, 2026. **(CAS Zone 1)**

- Contributed generalization analysis: derived the generalization bound of GNNs formed by monomial graph filters with Taylor expansion and asymptotic analysis.

2026.06 | F. Chen, **J. Zhang**, G. Liang, and Z. Lai, "A Design Principle for Graph Neural Networks," Pattern Recognition (PR), vol. 174, Art. no. 112999, 2026. **(CAS Zone 1)**
- Contributed the generalization theory: generalization bounds for graph-filter design of a family of GNNs via Rademacher complexity in a continuous-graph view.

2023.10 | Z. Lai, X. Chen, **J. Zhang**, H. Kong, and J. Wen, "Maximal Margin Support Vector Machine for Feature Representation and Classification," IEEE Trans. Cybern. (T-CYB), vol. 53, no. 10, pp. 6700-6713, 2023. **(CAS Zone 1)**
- Verified the optimization details: derivations and convergence of the joint feature-extraction and sparse-selection optimization.

2026.07 | F. Chen, Z. Lai, H. Lu, **J. Zhang**, Q. Wu, X. Luo, and H. Kong, "NSMNet: Stabilizing Linear State-space Memory for Breast Ultrasound Video Segmentation," ACM Multimedia (ACM-MM), 2026, Poster (accepted, to be published).
- Participated in idea formulation: identified rank vanishing issue of linear attention, designed stabilizing state-space memory for breast ultrasound video segmentation.

2026.07 | Q. Wu, Z. Lai, X. Luo, H. Kong, F. Chen, **J. Zhang**, and G. Liang, "A Large-Scale Multi-Modal Benchmark and Robust Text-Free Inference for Breast Ultrasound Video Segmentation," ACM Multimedia (ACM-MM), 2026, Poster (accepted, to be published).
- Led problem formulation: annotation challenges for multi-modal breast ultrasound video segmentation, formulated a semi-supervised scheme via text-free inference.

Research Projects

2020-2021 | Basic Experimental Project of the Student Innovation and Development Fund of Shenzhen University: **Robust Twin Bounded Support Vector Classifier with Manifold Regularization** (Sole Contributor)
- As the sole contributor, independently completed the full loop from proposal, algorithm design, implementation, experimental validation and project conclusion.
- Produced 1 IEEE T-CYB paper with open-sourced code (see publications); funded 10,000 CNY; successfully concluded.

2024-2025 | Graduate Independent Innovation Achievement Cultivation Project of Shenzhen University: **Efficient Big Data Algorithms Based on Kernel Methods** (First Contributor)
- As the first contributor, independently completed the full loop from proposal, algorithm design, implementation, experimental validation and project conclusion.
- Produced 2 papers (ICML 2025, Pattern Recognition 2025) with open-sourced code (see publications); funded 50,000 CNY; successfully concluded.

Teaching Experience

2026 Spring | **Multimedia Technology Fundamentals** (top-up bachelor's course), Independent Lecturer •School of Continuing Education, Shenzhen University
- **Independently taught the 12-week course** (design, slides, lecturing), covering data compression principles and MPEG/H.264/HEVC video coding.
- **Designed 2 hands-on labs and 3 comprehensive tests**, forming a complete "lecturing–assessment" closed loop.

2020-2025 | **SVM Special Topic** Lecturer (Machine Learning course) •Shenzhen University
- Delivered the SVM (Support Vector Machine) special topic in the Machine Learning course each year, with lecture notes and recorded videos (`SVM-Intro`).

2021-2023 | **Machine Learning** Course Teaching Assistant (undergraduate) •Shenzhen University

Skills

Programming | Python, C/C++, MATLAB; PyTorch, NumPy/SciPy, Scikit-learn; Git, Linux

Languages | Chinese (native); English: academic writing and communication

Academic Service

- Conference reviewing: ICME'24, AAAI'25, ICCV'25, CVPR'26, ICML'26, ACM-MM'26 (11 reviews in total)
- Journal reviewing: reviewer for TKDE, T-NNLS, T-PAMI (21 reviews completed)

Awards and Honors

University-level Awards / Social Scholarships

- 2021 Pengcheng Scholarship, Shenzhen University
- 2021, 2024 Scholarship of the Excellent PhD Candidate Selection and Funding Program, College of Computer Science and Software Engineering, Shenzhen University
- 2022 Excellent Undergraduate Graduate and Honorary Bachelor, Shenzhen University
- 2023 Scholarship of the Top-Talent Cultivation Program, Shenzhen University

Academic Competition Awards:

- 2020 Honorable Mention (Second Prize) in the Mathematical Contest in Modeling (MCM)
- 2020 Meritorious Winner (First Prize) in the Mathematical Modeling International Contest (China)
- 2021 Finalist (Outstanding Winner nomination, top 2% worldwide) in the Mathematical Contest in Modeling (MCM)